Power Budget

Team Number:	**105**
Project Name:	**RadioLink**
Team Member Names:	**Joseph, William, Mateo, Sultan**
Version:	**2**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors,

All Major Components	Component Name	Part Number	yVoltageRan	#	aximumCurre	alCurrent(mA)	Unit
	Op Amp	MCP6004	1.8–6.0 V	1	0.4	0.4	mA
	Electret Microphone	CMC-6027-24T	2–10 V	1	0.5	0.5	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.

+3.3V Power Rail	Component Name	Part Number	yVoltageRan	#	aximumCurre	alCurrent(mA)	Unit
	Op Amp	MCP6004	1.8–6.0 V	1	0.4	0.4	mA
	Electret Microphone	CMC-6027-24T	2–10 V	1	0.5	0.5	mA
						0	mA
						0	mA
	Subtotal					0.9	mA
	Safety Margin					25%	
	Total Current Required on +5V Rail					1.125	mA
c2. Regulator or Source Ch	PIC18F57Q43 Curiosity Nano VBUS	DM164150	3.3V	1	500	500	mA
	Total Remaining Current Available on +5V Rail					498.875	mA

C. For each power rail +3.3V Rail — PIC18F57Q43 Curiosity Nano VBUS (DM164150)

D. Select a specific

External Power Source 1	Component Name	Part Number	yVoltageRan	Outpu	aximumCurre	alCurrent(mA)	Unit
Power Source 1 Selection	DC Power Supply	N/A	5V	5V	500	500	mA
Power Rails Connected to External Power Source 1	+3.3V Rail — PIC18F57Q43 Curiosity Nano VBUS (DM164150)		3.3V				
	Total Remaining Current Available on External Power Source 1					499.27875	mA